

Seungheon Lee

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EDUCATION

	University of Seoul <i>Department of Mechanical Information Engineering</i> Bachelor Student, Class Rank: <u>1st</u> out of 46 students GPA: 4.15 / 4.5 (Major GPA: 4.35 / 4.5)	Seoul, Republic of Korea <i>Mar 2019 – Feb 2025</i>
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EXPERIENCE

	Hyundai Motor Company <i>Research Engineer</i> Managing the Robotaxi project, overseeing timelines and team coordination.	Hwasung, Republic of Korea <i>Jan 2025 – Present</i>
	Samsung Research <i>Software Engineer Intern</i> Developed monitoring system for LLM API servers: Implemented Kubernetes cluster and monitoring infrastructure using Prometheus and Grafana	Seoul, Republic of Korea <i>Jul 2024 – Aug 2024</i>
	Robotics Lab, University of Seoul <i>Undergraduate Researcher</i> Gained Research Experience and Participated in Various Competitions	Seoul, Republic of Korea <i>Mar 2022 – May 2024</i>

AREAS OF EXPERTISE

- Robotics: Autonomous Driving
- AI Applications: Generative AI, Machine Learning Quant

AWARDS AND SCHOLARSHIPS

Grand Prize (Minister of Science and ICT Award), 1st AI Semiconductor Technology Talent Selection Competition	<i>Dec 2024</i>
Academic Excellence Award, President of the University of Seoul	<i>Nov 2023</i>
1st place, RoboCup Autonomous Robot Manipulation Challenge, MathWorks, United States	<i>Jul 2023</i>
Grand Prize, 1st Autonomous Robot Race Competition, Autonomous Robot Race Competition Organizing Committee	<i>Apr 2023</i>
Best Presentation Award, Engineering Research Internship, University of Seoul Engineering Education Innovation Center	<i>Feb 2023</i>
Winner of Simulation Drone Contest, Korean Robotics Society	<i>May 2022</i>
OK Scholarship for College/Graduate Students, OK Scholarship Foundation	<i>Sep 2023 – Feb 2025</i>
Blue Lighthouse KB Dream Wave 2030 Promising Field Scholarship, Korea Scholarship Foundation	<i>Nov 2023 - Jun 2024</i>

CONFERENCES

- Minhoo Park, Seungheon Lee, Yonggun Lee, "Techniques for Creating Custom Article Illustrations with Stable Diffusion," KIBME 2024, Jeju, Korea – Oral *Jun 2024*
- Seungheon Lee, Taegyeom Lee, Ui Hun Sagong, Jung Hyun Choi, Won Bo Jang, Junyoung Lee, Murim Kim, Myeonjung Hwang, "Detection of Handle Grasping Pose Using a Deep Learning Based Grasping Pose Estimation Algorithm," KRoC 2024, Pyeongchang, Korea – Poster *Feb 2024*
- Seungheon Lee, Euihoon Sagong, Myeonjung Hwang, "Visualization of Singularity Distribution in Robot Manipulator using Monte Carlo Method," ICROS 2023, Samcheok, Korea – Poster *Jun 2023*

SKILLS

- Programming Languages - Python, C/C++
- Software/Tools - Linux, Kubernetes, MATLAB, PyTorch, Grafana, Weights & Biases (W&B)
- Hardware/Sensor - LiDAR(2D, 3D), GPS, RGB-D Camera

ACTIVITIES

Military Service

- Sergeant, Republic of Korea Army *Mar 2020 - Oct 2021*

Projects

- Computer Vision AI Model Optimization for NPU-based Inference: Participating in the 1st AI Semiconductor Competition, collaborating with Furiosa AI *Sep 2024 – Dec 2024*
- Creating Custom Article Illustrations with Stable Diffusion: project with KBS AI transformation team *Apr 2024 - Jun 2024*
- BartendAiRtist: Developed an AI-powered Android app for personalized cocktail recommendations and automated preparation using Chat GPT API *Mar 2023 - Dec 2023*
- RL-Enhanced-PurePursuit: Enhanced Pure Pursuit path-following algorithm using reinforcement learning with MATLAB's Reinforcement Learning Toolbox *Sep 2023 - Dec 2023*
- Zeus Industrial Robot ZERO Mission Challenge: focusing on tasks such as 6-axis articulated robot operation, robot vision, and point cloud processing, RoboWorld *Aug 2023 - Oct 2023*

Links

